

PROGRAM:

```
import java.io.*;
interface Mystack
{
    public void pop();
    public void push();
    public void display();
}
class Stack_array implements Mystack
{
    final static int n=5;
    int stack[]=new int[n];
    int top=-1;
    public void push()
    {
        try
        {
            BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
            if(top==(n-1))
            {
                System.out.println(" Stack Overflow");
                return;
            }
            else
            {
                System.out.println("Enter the element");
                int ele=Integer.parseInt(br.readLine());
                stack[++top]=ele;
            }
        }
        catch(IOException e)
        {
            System.out.println("e");
        }
    }
    public void pop()
    {
        if(top<0)
        {
            System.out.println("Stack underflow");
            return;
        }
        else
        {
            int popper=stack[top];
            top--;
            System.out.println("Popped element:" +popper); }
    }
}
```

```

public void display()
{
    if(top<0)
    {
        System.out.println("Stack is empty");
        return;
    }
    else
    {
        String str=" ";
        for(int i=0; i<=top; i++)
            str=str+" "+stack[i]+" <--";
        System.out.println("Elements are:"+str);
    }
}
}
}
class StackADT
{
    public static void main(String arg[])throws IOException
    {
        BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
        System.out.println("Implementation of Stack using Array");
        Stack_array stk=new Stack_array();
        int ch=0;
        do
        {
            System.out.println("1.Push 2.Pop 3.Display 4.Exit);
            System.out.println("Enter your choice:");
            ch=Integer.parseInt(br.readLine());
            switch(ch)
            {
                case 1:
                    stk.push();
                    break;
                case 2:
                    stk.pop();
                    break;
                case 3:
                    stk.display();
                    break;
                case 4:
                    System.exit(0);
            }
        }
        while(ch<5);
    }
}

```